elements. In mathematical terms, the priority is a univocal mapping between elements and a number, usually with the convention that lower values mean higher priority.

Some of these definitions for priority are so common that we categorize them as separate data structures. Let’s see a few of them.

D.1 Bag

A bag is a collection that only support the add and iterate operations. It's not possible to remove items from a bag. After checking if the container is actually empty, clients can iterate through the elements; the actual order, however, is unspecified by definition, and clients must not rely on it.

Bags are useful when you need to collect objects and process them as a whole set rather than individually. For instance, you could collect samples and then, at a later point, compute statistics on them, like average or standard deviation - the order is of course irrelevant in that case.

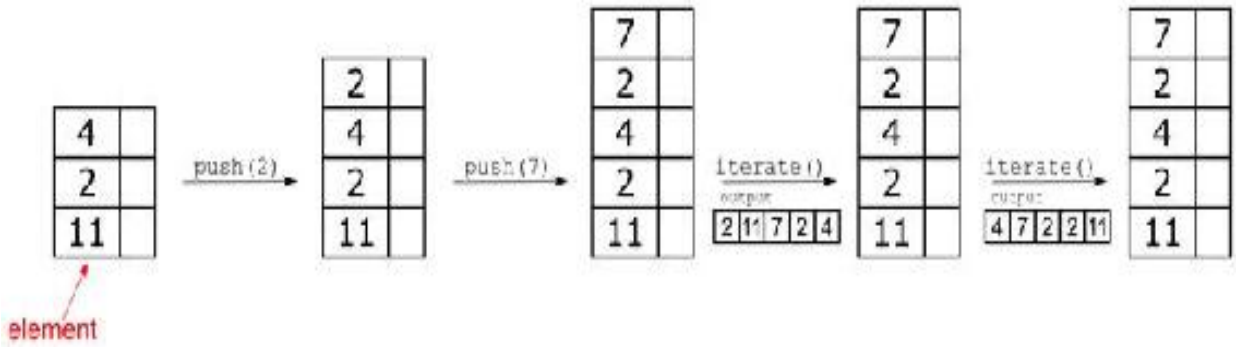


Figure D.1 Operations on a bag. Insertion adds element to the container, but sets no index for them.